



Training Feed-Forward Multi-Layer Perceptron Artificial Neural Networks with a Tree-Seed Algorithm

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Received: 28 March 2020 / Accepted: 13 August 2020 / Published online: 2 September 2020
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Abstract

The artificial neural network (ANN) is the most popular research area in neural computing. A multi-layer perceptron (MLP) is an ANN that has hidden layers. Feed-forward (FF) ANN is used for classification and regression commonly. Training of FF MLP ANN is performed by backpropagation (BP) algorithm generally. The main disadvantage of BP is trapping into local minima. Nature-inspired optimizers have some mechanisms escaping from the local minima. Tree-seed algorithm (TSA) is an effective population-based swarm intelligence algorithm. TSA mimics the relationship between trees and their seeds. The exploration and exploitation are controlled by search tendency which is a peculiar parameter of TSA. In this work, we train FF MLP ANN for the first time. TSA is compared with particle swarm optimization, gray wolf optimizer, genetic algorithm, ant colony optimization, evolution strategy, population-based incremental learning, artificial bee colony, and biogeography-based optimization. The experimental results show that TSA is the best in terms of mean classification rates and outperformed the opponents on 18 problems.

Keywords Tree-seed algorithm · Multi-layer perceptron · Training neural network · Artificial neural network · Neural networks · Nature inspired algorithms

1 Introduction

Neural computing mimics the human brain which is the most complex organ of the human body [1]. Neural networks simulate the connections in the human brain [2]. This simulation is named as the artificial neural network (ANN). Basically ANN takes inputs, computes them, and produces the outputs. This process is a learning process. Learning has two main types: supervised and unsupervised. In supervised learning, the training data have output labels but in unsupervised learning, the data have not got output labels. ANN is a balancer between the input and outputs. In the literature, there are various types of networks such as feed-forward (FF) [3], Kohonen [4], radial-basis function (RBF) [5], recurrent neural [6], spiking neural [7]. FF is a network that has one way (one direction). In FF, the association of inputs and outputs is provided with weights and biases. If a FF ANN has hidden layers, it is named as multi-layer perceptron (MLP) [8, 9]. MLP has three layers: input, hidden, and output. Training data are used for learning the hidden weights between the attributes and class labels. The deterministic and stochastic learning approaches are used for training an ANN. The gradient-based methods and backpropagation (BP) algorithm are deterministic methods [10]. If the training data does not change, deterministic methods produce the same results. The deterministic methods are simple and speedy. Stochastic methods try to improve the learning rate during the iterations. Thus, time usage is higher than the deterministic methods, but it gives us better results. The main drawback of the deterministic method is the initial solutions dependency. Stochastic optimization techniques start with random solutions. These random solutions are evolved in every iteration, and the main advantage is avoiding the local optima. Nature-inspired optimizers are in the stochastic optimization techniques group. Most of these methods are multi-solution-based algorithms but some of them like hill climbing [11] and simulated annealing [12, 13] are single solution-based algorithms. TSA, particle swarm optimization (PSO), gray wolf optimizer (GWO), genetic algorithm (GA), ant colony optimization (ACO), evolution strategy (ES), population-based incremental learning (PBIL), arti-

tron (MLP) [8, 9]. MLP has three layers: input, hidden, and output. Training data are used for learning the hidden weights between the attributes and class labels. The deterministic and stochastic learning approaches are used for training an ANN. The gradient-based methods and backpropagation (BP) algorithm are deterministic methods [10]. If the training data does not change, deterministic methods produce the same results. The deterministic methods are simple and speedy. Stochastic methods try to improve the learning rate during the iterations. Thus, time usage is higher than the deterministic methods, but it gives us better results. The main drawback of the deterministic method is the initial solutions dependency. Stochastic optimization techniques start with random solutions. These random solutions are evolved in every iteration, and the main advantage is avoiding the local optima. Nature-inspired optimizers are in the stochastic optimization techniques group. Most of these methods are multi-solution-based algorithms but some of them like hill climbing [11] and simulated annealing [12, 13] are single solution-based algorithms. TSA, particle swarm optimization (PSO), gray wolf optimizer (GWO), genetic algorithm (GA), ant colony optimization (ACO), evolution strategy (ES), population-based incremental learning (PBIL), arti-

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ficial bee colony (ABC), biogeography-based optimization (BBO) are some of the multi-solution nature-inspired optimizers. These algorithms are not only used in training of FF ANN MLPs but also used in various applications like feed formulation [14], traveling salesman problem [15], layout problem [16], and combinatorial problems [17]. Also, ANN is used in many applications like forecasting [18], classification [19, 20], estimation [21, 22], and prediction [23]. The ANN is not only used for classifying stationary signals but also non-stationary signals [24]. Various techniques are used in classification of non-stationary signals, for example, Koh and Woo [24] combined ensemble technique and multi-view learning for classifying stationary signals; Boashash and Ouelha [25] focused on extracting information from non-stationary signals; Delsy et al. [26] extracted the features from non-stationary signals and classified them with backpropagation network. According to the No Free Lunch (NFL) [27] theorem, a nature-inspired optimizer cannot solve all optimization problems successfully. Therefore, since the GA is proposed in 1975, more than 300 nature-inspired algorithms are proposed until now. Every nature-inspired optimizer has peculiar property, so, in this work, we want to prove the success of the TSA to training MLP. TSA is an effective solver on low dimensional problems. In this work, we modify the basic TSA for solving large-scale MLP training.

The remainder of the paper is organized as follows: Sect. 1.1 gives the main contribution of the study. In Sect. 2, the related works are given. FF MLP ANN and TSA are examined in Sects. 3.1 and 3.2, respectively. The experimental setup and information about datasets are given in Sect. 4. The results and discussion are located in Sect. 5. Finally, in Sect. 6 we conclude the work.

1.1 The Main Contribution of the Study

- TSA is used for training the FF MLP ANN for the first time.
- TSA is compared and outperformed on 18 different datasets (6 to 6786 dimensions, 4 to 7400 samples) with 8 metaheuristic algorithms.
- TSA finds eligible weights and biases of FF MLP ANN.
- The parameter adjustment for the basic TSA has increased the mean classification accuracy.
- TSA is the best solver on 18 different type datasets in terms of mean classification rates.

2 Literature Review

Neural computing and nature-inspired optimizers are a huge research domain. Training feed-forward multi-layer perceptron artificial neural networks subject is not a fresh idea, but

it is a most discussed, alive, and growing research problem in the literature. Therefore, in this section, we only focus on the recent applications related to the training of FF MLP ANN with nature-inspired optimizers and the literature of TSA.

Wienholt [28] uses ES for minimizing the system error of an MLP. Seiffer [29] propose a GA approach for avoiding local minima on training MLP in 2001. Mendes et al. [30] use PSO for training MLPs on classification and regression tasks in 2002. In 2005, Blum and Socha [31] extend ACO for pattern classification on medical data. Karaboga et al. [32] use ABC for training FF ANN. Five function approximation problems are used in experiments. ABC outperforms BP and GA in this work. Mirjalili et al. [33] hybridize PSO and gravitational search algorithm, and it is named PSOGSA. In this work, MLP is trained with PSOGSA. The obtained results are compared with PSO and GSA. PSOGSA is better than PSO and GSA in terms of convergence, training error, and classification rate. Mirjalili et al. [34] train MLP with BBO in 2014. Five classification and six approximation datasets are used for experiments. BBO is outperformed to PSO, GA, ACO, ES, and PBIL. Also, the obtained results are compared with the BP algorithm and extreme learning machine. Mirjalili [8] investigates the effectiveness of the GWO on training MLP in 2015. Five classification and three function approximation datasets are used for determining the performance of GWO. GWO creates better results than PSO, GA, ACO, ES, and PBIL. Amirsadri et al. [35] combine BP and GWO for training MLP. Levy flight technique is used to improve the exploration capability of GWO like [36]. BP increases the exploitation capability of GWO. The success of the proposed model is shown on 12 classification and function-approximation datasets. Xu et al. [37] modify ABC with the global best-guided approach for continuous optimization problems, and this method is named ABC-ISB. ABC-ISB is compared with ABC variants in the literature. In this work, basic ABC and ABC-ISB is compared on training MLP. ABC-ISB creates promising results. Zhang et al. [38] optimize the weights and biases of a MLP with improved GWO, and their approach is named RSMGWO in 2019. RSMGWO used a random opposition learning strategy for avoiding the local optima. Nineteen different cancer-related datasets are used for experiments. RSMGWO produces competitive results. Heidari et al. [39] use ant–lion optimizer for the training MLP in 2020, and their approach is named as ALOMLP. ALOMLP outperforms GA, PBIL, DE, and PSO in this work. Dalwinder et al. [40] weighted the features of the datasets, for increasing the classification rate in 2020. In this work, an ant–lion optimizer is used for training the MLPs. Three breast cancer datasets are used in the experimental setup. The obtained results showed that this paradigm increases the classification rate. Faris et al. [41] train MLP with multi-verse optimizer (MVO). Nine different



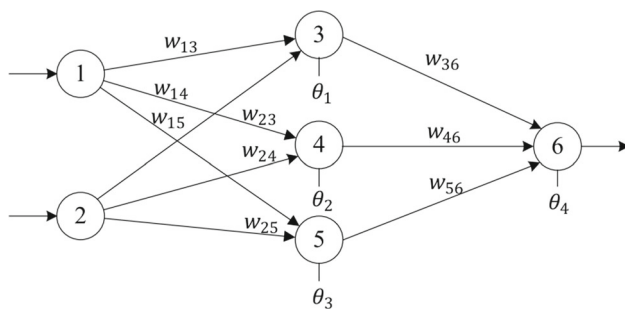


Fig. 1 The structure of 2–3–1 MLP

bio-medical datasets selected from the UCI machine learning repository are used in experiments. MVO is compared with GA, PSO, DE, firefly, and cuckoo search algorithms. The experimental results show that MVO produces compatible results. The metaheuristic algorithms are not used only for time-series prediction but also used in image processing [42], electrical machine design [43], and optimization of energy consumption in wireless sensor network [44].

TSA is another iterative continuous search algorithm proposed by Kiran [45] in 2015. In the literature, TSA is used in the wide range of research areas such as constrained versions of TSA [46, 47], engineering optimization problems solved with TSA [48–58], RBF network training and applications with TSA [48, 59], parallel versions of TSA [60, 61], image processing with TSA [62–65], binary optimization with TSA [66–68], improved versions of TSA [50, 69–76], feature selection with TSA [77], discrete versions of TSA [15].

Until now, there is no work in the literature for training MLP with TSA; our main motivation is to present the effectiveness of TSA for training MLPs.

3 Materials and Methods

3.1 Feed-Forward Neural Network and Multi-Layer Perceptron

The FF neural network is a neural network that has one way (one direction) between their neurons. If a NN has hidden layers, it is named as MLP. In this study, vector representation is used for individuals. The individual for the 2-3-1 MLP that is presented in Fig. 1 is $X = [W_{13} W_{23} W_{14} W_{24} W_{15} W_{25} W_{36} W_{46} W_{56} \theta_1 \theta_2 \theta_3 \theta_4]$. The dimension is calculated as $((\text{InputNumber} + \text{OutputNumber} + 1) * \text{HiddenNodesNumber}) + 1$.

Mean square error (MSE) for all training samples is used as the objective function. Equation 1 shows this calculation:

$$\text{MSE} = \sum_{t=1}^T \frac{\sum_{i=1}^m (R_i^t - C_i^t)^2}{T} \quad (1)$$

where T is the number of training samples, m is the number of outputs, C_i^t is the created output value for i th input for t th training sample, R_i^t is the real output value for i th input for t th training sample.

3.2 Tree-Seed Algorithm

TSA was proposed by Kiran [45] for solving unconstrained continuous optimization problems in 2015. TSA simulates the relationship between trees and their seeds. TSA is a population-based swarm intelligence techniques. It has two peculiar parameters. These parameters are the Search Tendency (ST) and the Number of Seeds (NS). ST controls the seed creation direction. The population is named as stand in TSA. Kiran [45] recommends that the NS can be between 10% of the stand size and 25% of the stand. But if necessary one can change this number. In TSA, the trees and seeds correspond to the possible solution of an optimization problem. At the initialization phase, the population is created randomly in a predetermined search space. The trees and seeds are D -dimensional vectors, and D is the dimensionality of an optimization problem. The search process is a trade-off of exploration and exploitation. If this trade-off is balanced, the algorithm creates more qualified solutions. In TSA, this situation is controlled by the ST parameter with two different seed creation formulas given in Eqs. 2 and 3, respectively.

$$\text{Seed}_{(k,j)} = \text{Tree}_{(i,j)} + (\text{Best}_j - \text{Tree}_{(r,j)}) \times \text{Rand}(-1, 1) \quad (2)$$

$$\text{Seed}_{(k,j)} = \text{Tree}_{(i,j)} + (\text{Tree}_{(i,j)} - \text{Tree}_{(r,j)}) \times \text{Rand}(-1, 1) \quad (3)$$

where k is the index of the seed, j is the index of the dimension, r is the index of the random neighbor tree, Best is the best tree obtained so far, $\text{Rand}(-1, 1)$ is a random number between -1 and 1 . The Equation 2 provides the exploitation and Equation 3 provides the exploration. The detailed pseudocode of the basic TSA is given in Fig. 2.

3.3 Training MLP with TSA

This section describes how to train a FF MLP with TSA deeply. TSA is a continuous optimization algorithm, and Sect. 3.2 gives detailed information about TSA. The main aim is to determine the optimum parameters of MLP. These



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Determine the number of trees (N)
Determine the search tendency (ST) parameter
Determine the maximum function evaluation number (Maxfes)
D is the dimensionality of the problem
Initialize the trees
Evaluate the trees
Fes=N
WHILE Fes<Maxfes
    FOR i=1 to N
        Determine the number of seeds between the 10% of the population size and 25%of the
        population (NS)
        Select a random neighbor tree (r) that not equals the current tree
        FOR k=1 to NS
            FOR j=1 to D
                IF rand<ST
                    Seed(k,j)=Tree(i,j)+rand(-1,1)*(Best(j)-Tree(r,j))
                    Relocate the seeds if cross the search space boundaries
                ELSE
                    Seed(k,j)=Tree(i,j)+rand(-1,1)*(Tree(i,j)-Tree(r,j))
                    Relocate the seeds if cross the search space boundaries
                END
            END
        END
        Determine the best seed with a greedy selection mechanism
        If the best seed is better than its tree, then the tree is removed from the search space and the
        best seed become a tree
    END
    Determine the best tree with a greedy selection mechanism
END

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Fig. 2 The detailed pseudocode of the basic TSA

parameters are clearly explained in Sect. 3.1. At the initialization phase, these values are started as a random vector. After that, this vector is optimized by TSA, and finally optimized parameters of a MLP are produced by TSA. The flowchart of the proposed method is given in Fig. 3.

3.4 The Computational Complexity of the Proposed Method

The computational complexity of the proposed method is related to the structure of the MLP, the number of instances in the training data, the stand size, the number of maximum

function evaluation numbers, and the number of seeds. The Big-O notation of the computational complexity of the proposed method is given in Eq. 4.

$$O(\text{TSA}, \text{MLP}) = O(\text{Maxfes}(O(\text{MLP}) + O(\text{TSA}))) \quad (4)$$

where Maxfes is the number of maximum function evaluation numbers, $O(\text{MLP})$ is the Big-O notation of MLP, and it is calculated as in Eq. 5, $O(\text{TSA})$ is the Big-O notation of TSA, and it is calculated as in Eq. 6.

$$O(\text{MLP}) = (t(h + o)) \quad (5)$$



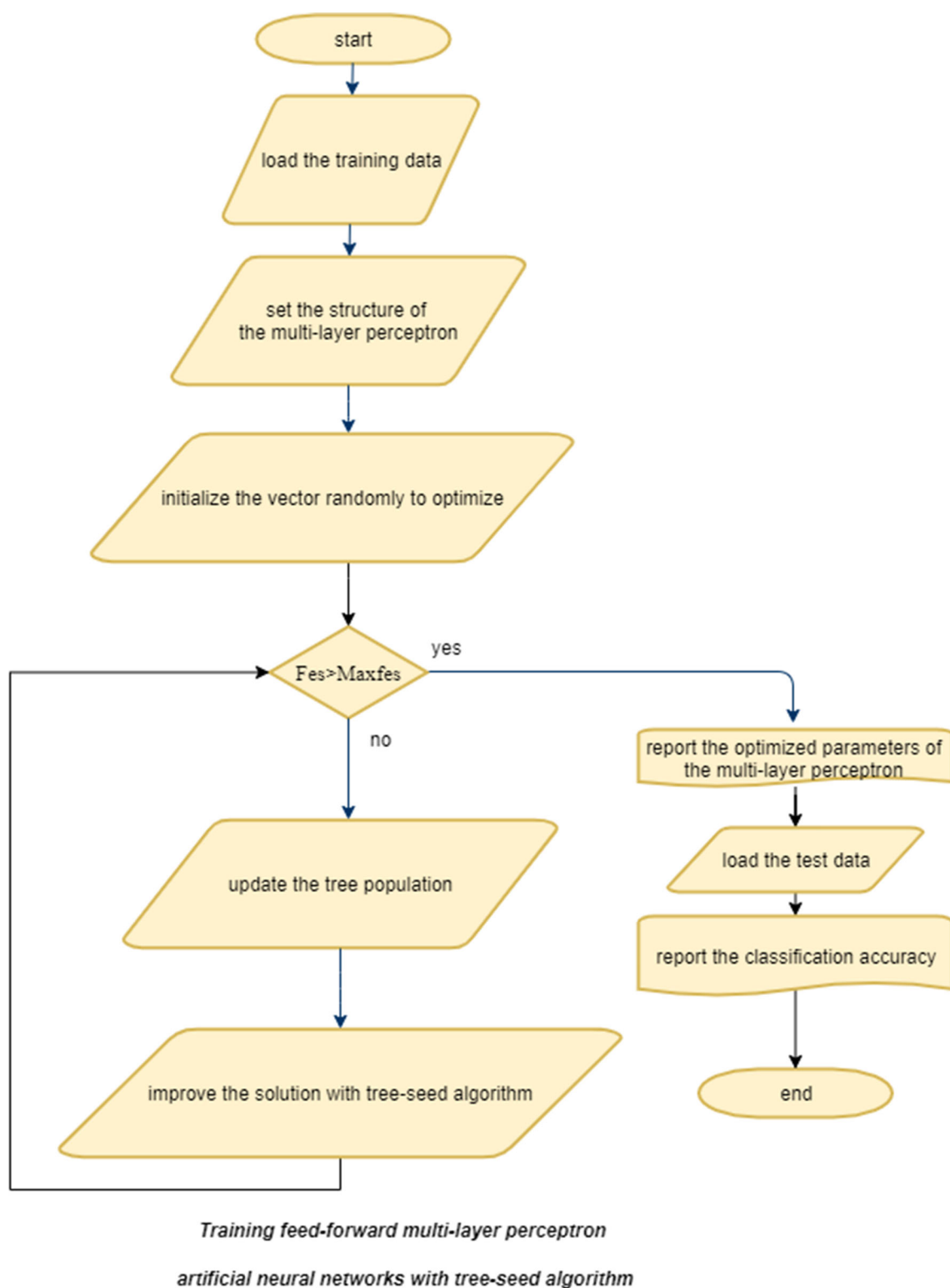


Fig. 3 The flowchart of the proposed method



Table 1 The details of datasets

No	Dataset name	Number of attributes	MLP structure	Dimensions	Weight numbers	Bias numbers	Range
1	XOR6	2	2-2-1	6	6 weights	0 bias	[− 100,100]
2	XOR9	2	2-2-1	9	6 weights	3 biases	[− 10,10]
3	XOR13	2	2-3-1	13	9 weights	4 biases	[− 10,10]
4	3-bit Parity	3	3-3-1	16	12 weights	4 biases	[− 10,10]
5	4-bit Enc. Dec	4	4-2-4	22	16 weights	6 biases	[− 10,10]
6	3-bits XOR	3	3-7-1	36	28 weights	8 biases	[− 10,10]
7	Sigmoid	1	1-15-1	46	30 weights	16 biases	[− 10,10]
8	Cosine	1	1-15-1	46	31 weights	17 biases	[− 10,10]
9	Sine	1	1-15-1	46	32 weights	18 biases	[− 10,10]
10	Balloon	4	4-9-1	55	45 weights	10 biases	[− 10,10]
11	Iris	4	4-9-3	75	63 weights	12 biases	[− 10,10]
12	Breast Cancer	9	9-19-1	210	190 weights	20 biases	[− 10,10]
13	Heart	22	22-45-1	1082	1035 weights	46 biases	[− 10,10]
14	Banknote	4	4-9-1	55	45 weights	10 biases	[− 10,10]
15	Diabetic	19	19-39-1	820	780 weights	40 biases	[− 10,10]
16	Twonorm	20	20-41-1	903	861 weights	42 biases	[− 10,10]
17	Ringnorm	20	20-41-1	903	861 weights	42 biases	[− 10,10]
18	Spambase	57	57-115-1	6786	6670 weights	116 biases	[− 10,10]

where t is the number of instances in the training data, h is the number of hidden nodes in the MLP, o is the number of output values. In this work, h and o are smaller than t , so in the worst-case $O(\text{MLP}) = t$.

$$O(\text{TSA}) = (N \times NS \times D) \quad (6)$$

where N is the stand size, NS is the number of seeds, D is the dimensionality of the training dataset. In the best case $NS = N/10$ and in the worst-case $NS = N/4$. NS must be smaller than N . Generally D is smaller than t ; for easing the calculation we suppose $D = t$. So, in the worst-case $O(\text{TSA}) = N \times N/4 \times t$.

The overall computational complexity of the proposed method is given in Eq. 7.

$$O(\text{TSA}, \text{MLP}) = O(\text{Maxfes}(t + N \times N/4 \times t)) \quad (7)$$

4 Experimental Setup

In this section, we gave the details of our experimental setup. The details of the datasets (the name of the dataset, the number of the attributes of the dataset, the MLP structure for training the dataset, the dimension of the dataset, the total weight numbers of the dataset, the total bias numbers of the dataset, and the search space range for the dataset) are given in Table 1. Determining the performance of the algorithm, 18 different datasets (XOR6, XOR9, XOR13, 3-bit

Parity, 4-bit Encoder Decoder, 3-bits XOR, Sigmoid, Cosine, Sine, Balloon, Iris, Breast Cancer, Heart, Banknote, Diabetic, Twonorm, Ringnorm, and Spambase) are used in experiments. The large datasets have more than 1000 training/test samples discussed in Sect. 5.3. There is no strict rule for selecting the number of hidden nodes. Equation 8. is used for the determining the number of hidden nodes.

$$H = 2 \times I + 1 \quad (8)$$

where H is the number of hidden nodes of MLP and I is the number of input nodes. For function approximation datasets, the number of hidden nodes is set as 15.

In this work, we use nine algorithms. All specific parameters of these algorithms are listed in Table 2.

The maximum iteration numbers, the maximum number of the evaluation numbers, the population sizes (for PSO, GWO, GA, ACO, ES, PBIL, and BBO), the stand size of TSA, and the colony size of ABC for every dataset are given in Table 3.

The information about training/test samples is given in Table 4.

The datasets are mapped to $[-1, +1]$ space with the min–max normalization method that was formulated as seen in Eq. 5.

$$X' = \frac{(X - x_{\min}) \times (1 - (-1))}{(x_{\max} - x_{\min})} + (-1) \quad (9)$$

Table 2 Specific parameters of the algorithms that used in this work

Algorithm	Parameter	Value
TSA	Search Tendency	0.1
	Number of Seeds N: Stand Size	$N*0.1—N*0.25$
GWO	a (linearly decreased)	2 to 0
ABC	Colony Size (CS) N: Population Size	$N/2$
	Limit D: Dimension of the problem	$CS*D$
BBO	Habitat modification probability	1
	Immigration probability bounds per gene	[0,1]
	Step size for numerical integration of probabilities	1
	Max immigration (I) and Max emigration (E)	1
PSO	Mutation probability	0.005
	Cognitive constant (C1)	1
	Social constant (C2)	1
	Inertia constant (w)	0.3
GA (real coded, selection Roulette wheel)	Crossover single point (probability)	1
	Mutation uniform (probability)	0.01
ACO	Initial pheromone (s0)	$1.00E-06$
	Pheromone update constant (Q)	20
	Pheromone constant (q0)	1
	Global pheromone decay rate (pg)	0.9
	Local pheromone decay rate (pt)	0.5
	Pheromone sensitivity (a)	1
	Visibility sensitivity (b)	5
ES	Lambda	10
	Sigma	1
PBIL	Learning rate	0.05
	Good population member	1
	Bad population member	0
	Elitism parameter	1
	Mutational probability	0.1

where X' is the mapped value, X is the real value, $x_{\max}$ is the maximum value of the dataset, $x_{\min}$ is the minimum value of the dataset.

4.1 Balloon Dataset

The balloon dataset, about blowing up a balloon, has 4 attributes (color, size, act, and age) and 20 training/test samples (4 repeated). If the balloon is inflated, the output is 1; otherwise, the output is zero. The string input variables are converted to binary format. The color values are yellow and purple, the size values are small and large, the act values are stretch and dip, and the age values are adult and child. The files related to the dataset can be found in <https://archive.ics.uci.edu/ml/datasets/Balloons>.

4.2 Iris Dataset

The Iris dataset, about class of iris plant, has 4 attributes (sepal length, sepal width, petal length, and petal width) and 150 training/test samples. If the class is Iris Setosa the output is -1 , if the class is Iris Versicolour the output is 0, and if the class is Iris Virginica the output is 1. The input variables are mapped between -1 and 1 with the min–max normalization method that aforementioned before. The files related to the dataset can be found in <https://archive.ics.uci.edu/ml/datasets/Iris>.

4.3 Breast Cancer Dataset

The Breast cancer dataset, about patients who cancer or not, has 10 attributes (id, clump thickness, uniformity of cell size, uniformity of cell shape, marginal adhesion, single epithelial cell size, bare nuclei, bland chromatin, normal nucleoli, and mitoses) and 599 training/100 test samples. If the cancer is benign the output is 0; if the cancer is malignant, the output is 1. The input variables are converted to continuous variables between -1 and 1 with the min–max normalization method that aforementioned before. The files related to the dataset can be found in [https://archive.ics.uci.edu/ml/datasets/breast+cancer+wisconsin+\(original\)](https://archive.ics.uci.edu/ml/datasets/breast+cancer+wisconsin+(original)).

4.4 Heart Dataset

The Heart dataset, about patients who has heart disease or not, has 22 attributes (binary features that extracted from images) and 267 training/test samples. In this work, we only use the first 80 training/test samples. If the patient is normal, the output is 0; if the patient is abnormal, the output is 1. The files related to the dataset can be found in <https://archive.ics.uci.edu/ml/datasets/spect+heart>.



Table 3 The population/colony/stand sizes and iteration/function evaluation numbers

No	Dataset name	Population size	Maximum iteration number	Maximum function evaluations	Colony size	Stand size
1	XOR6	50	250	12,500	25	10
2	XOR9	50	250	12,500	25	10
3	XOR13	50	250	12,500	25	10
4	3-bit Parity	50	250	12,500	25	10
5	4-bit Enc. Dec	50	250	12,500	25	10
6	3-bits XOR	50	250	12,500	25	10
7	Sigmoid	200	250	50,000	100	50
8	Cosine	200	250	50,000	100	50
9	Sine	200	250	50,000	100	50
10	Balloon	50	250	12,500	25	10
11	Iris	200	250	50,000	100	50
12	Breast cancer	200	250	50,000	100	50
13	Heart	200	250	50,000	100	50
14	Banknote	50	250	12,500	25	10
15	Diabetic	50	250	12,500	25	10
16	Twonorm	50	250	12,500	25	10
17	Ringnorm	50	250	12,500	25	10
18	Spambase	50	250	12,500	25	10

Table 4 The training/test samples information

No	Dataset name	Training samples	Test samples	NOTrS	NOTeS
1	XOR6	(0 0;0 1;1 0;1 1) = >(0;1;1;0)	Same as training samples	4	4
2	XOR9	(0 0;0 1;1 0;1 1) = >(0;1;1;0)	Same as training samples	4	4
3	XOR13	(0 0;0 1;1 0;1 1) = >(0;1;1;0)	Same as training samples	4	4
4	3-bit Parity	(0 0 0;0 0 1;0 1 0;0 1 1;1 0 0;1 0 1;1 1 0;1 1 1) → (0;1;1;0;1;0;0;1)	Same as training samples	8	8
5	4-bit Enc. Dec	(0 0 0 1;0 0 1 0;0 1 0 0;1 0 0 0) → (0 0 0 1;0 0 1 0;0 1 0 0;1 0 0 0)	Same as training samples	4	4
6	3-bits XOR	(0 0 0;0 0 1;0 1 0;0 1 1;1 0 0;1 0 1;1 1 0;1 1 1) → (0;1;1;0;1;0;0;1)	Same as training samples	8	8
7	Sigmoid	x in [− 3;0.1;3]	x in [− 3;0.05;3]	61	121
8	Cosine	x in [1.25;0.05;2.75]	x in [1.25;0.04;2.75]	31	38
9	Sine	x in [− 2π;0.1;2π]	x in [− 2π;0.05;2π]	126	252
10	Balloon	The details are given in Sect. 4.1	Same as training samples	20	20
11	Iris	The details are given in Sect. 4.2	Same as training samples	150	150
12	Breast cancer	The details are given in Sect. 4.3	Not the same as training samples	599	100
13	Heart	The details are given in Sect. 4.4	Same as training samples	80	80
14	Banknote	The details are given in Sect. 4.5	Same as training samples	1372	1372
15	Diabetic	The details are given in Sect. 4.6	Same as training samples	1151	1151
16	Twonorm	The details are given in Sect. 4.7	Same as training samples	7400	7400
17	Ringnorm	The details are given in Sect. 4.8	Same as training samples	7400	7400
18	Spambase	The details are given in Sect. 4.9	Same as training samples	4601	4601

NOTrS: Number of training samples NOTeS: Number of test samples



Table 5 Experimental results of the XOR6 dataset

XOR6	ABC	ACO	BBO	ES	GA	GWO	PBIL	PSO	TSA
Mean	1.98E−23	2.40E−23	2.97E−29	6.73E−29	2.34E−29	1.05E−29	1.24E−29	1.05E−29	1.05E−29
Best	1.06E−29	1.24E−29	2.97E−29	1.06E−29	1.24E−29	1.03E−29	1.24E−29	1.03E−29	1.03E−29
Worst	3.99E−22	7.16E−22	2.97E−29	5.48E−28	8.41E−29	1.09E−29	1.24E−29	1.09E−29	1.09E−29
SD	7.39E−23	1.31E−22	2.28E−44	1.16E−28	2.40E−29	1.66E−31	2.85E−45	1.91E−31	1.88E−31
Median	5.10E−26	8.29E−29	2.97E−29	2.51E−29	1.24E−29	1.05E−29	1.24E−29	1.05E−29	1.05E−29
Mean time	1.3564	1.6653	1.3946	1.4546	1.8817	1.4412	0.7653	2.2254	2.5986
Friedman rank	8.4	7.8	6.6	5.8	5.5	1.9	4.7	2.3	1.9
Manuel rank	2	3	4	2	3	1	3	1	1
Wilcoxon	1.92E−06	1.73E−06	1.73E−06	2.13E−06	1.73E−06	4.91E−01	1.73E−06	2.06E−01	0.00E+00
Classification rate (%)	100	100	100	100	100	100	100	100	100

4.5 Banknote Authentication Dataset

The Banknote authentication dataset is related to whether the banknote is valid or invalid. It has four continuous attributes (variance of wavelet transformed image, skewness of wavelet transformed image, curtosis of wavelet transformed image, the entropy of image), and 1372 training/test samples. The files related to the dataset can be found in <https://archive.ics.uci.edu/ml/datasets/banknote+authentication>.

4.6 Diabetic Retinopathy Debrecen Dataset

The Diabetic Retinopathy Debrecen dataset contains information about people who have diabetic retinopathy or not. It has 19 continuous and integer attributes and 1151 training/test samples. The files related to the dataset can be found in <https://archive.ics.uci.edu/ml/datasets/Diabetic+Retinopathy+Debrecen+Data+Set>.

4.7 Twonorm Dataset

The Twonorm dataset is an artificial dataset that has 20 continuous attributes and 7400 training/test samples. The files related to the dataset can be found in <https://www.cs.toronto.edu/~delve/data/twonorm/desc.html>

4.8 Ringnorm Dataset

The Ringnorm dataset is an artificial dataset that has 20 continuous attributes and 7400 training/test samples. The files related to the dataset can be found in <https://www.cs.toronto.edu/~delve/data/ringnorm/desc.html>.

4.9 Spambase Dataset

The Spambase dataset is about the classifying emails as spam or not. It has 57 continuous or integer attributes and 4601

training/test samples. The files related to the dataset can be found in <https://archive.ics.uci.edu/ml/datasets/Spambase/>.

5 Results and Discussion

All obtained results and discussion about these results are located in this section. The best training results and the best classification rates are highlighted with bold type text and italic background in Tables 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 21, 22, 23, 24. The statistical tests are important for determining the significant difference between the obtained results. In this work, two different statistical tests are conducted. These are the Wilcoxon signed rank test and Friedman's test. The 30 runs obtained results are used in these tests. The significance level is taken as 5% (0.05), and the p values of the Wilcoxon signed rank test and mean rank values of Friedman's test are located in Tables 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17 and Tables 21, 22, 23, 24. The large datasets have more than 1000 training/test samples discussed in Sect. 5.3.

The experimental results for the XOR6 dataset are given in Table 5. TSA, PSO, and GWO share the first position in terms of mean training error results. The classification rate is 100% for all methods. Therefore, XOR6 is not an identifier problem.

The experimental results for the XOR9 dataset are given in Table 6. ABC, BBO, GA, PBIL, and GWO share the first position in terms of mean training error results. The classification rate is 100% for all methods. Therefore, XOR9 is not an identifier problem.

The experimental results for the XOR13 dataset are given in Table 7. ABC, BBO, GA, and GWO share the first position in terms of mean training error results. The classification rate is 100% for all methods. Therefore, XOR13 is not an identifier problem.



Table 6 Experimental results of the XOR9 dataset

XOR9	ABC	ACO	BBO	ES	GA	GWO	PBIL	PSO	TSA
Mean	2.70E-09	2.30E-06	2.69E-09	1.86E-05	2.69E-09	2.89E-06	3.23E-09	1.75E-08	2.94E-09
Best	2.69E-09	1.74E-08	2.69E-09	3.11E-09	2.69E-09	2.69E-09	2.69E-09	2.70E-09	2.70E-09
Worst	2.71E-09	2.58E-05	2.69E-09	0.00028	2.69E-09	5.12E-05	5.75E-09	1.15E-07	3.67E-09
SD	3.03E-12	4.84E-06	4.21E-25	5.34E-05	1.08E-24	1.12E-05	8.34E-10	2.41E-08	2.60E-10
Median	2.69E-09	8.89E-07	2.69E-09	7.10E-07	2.69E-09	2.70E-09	2.69E-09	9.30E-09	2.83E-09
Mean time	1.4393	1.4955	1.4015	1.4390	1.8169	1.4570	0.7771	2.2102	2.7048
Friedman rank	3.9	8.4	1.7	8.2	1.7	4.6	4.0	6.8	5.6
Manuel rank	1	4	1	3	1	1	1	2	2
Wilcoxon	1.73E-06	1.73E-06	1.73E-06	1.73E-06	1.73E-06	6.16E-04	6.58E-01	4.73E-06	0.00E+00
Classification rate (%)	100	100	100	100	100	100	100	100	100

Table 7 Experimental results of the XOR13 dataset

XOR13	ABC	ACO	BBO	ES	GA	GWO	PBIL	PSO	TSA
Mean	1.43E-13	2.20E-07	1.40E-13	1.12E-06	1.73E-09	4.51E-08	5.95E-10	1.32E-09	6.72E-10
Best	1.40E-13	4.49E-11	1.40E-13	6.51E-10	1.40E-13	1.40E-13	1.52E-13	2.54E-13	1.55E-13
Worst	1.59E-13	1.72E-06	1.40E-13	2.24E-05	2.16E-08	6.81E-07	3.96E-09	2.32E-08	5.21E-09
SD	4.41E-15	3.82E-07	2.57E-29	4.15E-06	4.17E-09	1.61E-07	1.03E-09	4.57E-09	1.50E-09
Median	1.41E-13	5.03E-08	1.40E-13	2.28E-08	1.40E-13	1.42E-13	6.08E-13	1.34E-11	3.92E-13
Mean time	2.0267	2.1107	2.0286	1.8804	2.4000	2.0502	0.9473	2.7743	3.7753
Friedman rank	2.8	8.3	1.4	8.1	3.5	4.6	5.2	5.8	5.3
Manuel rank	1	5	1	6	1	1	2	4	3
Wilcoxon	1.73E-06	2.60E-06	1.73E-06	3.88E-06	5.30E-01	9.26E-01	8.13E-01	1.92E-01	0.00E+00
Classification rate (%)	100	100	100	100	100	100	100	100	100

Table 8 Experimental results of the 3-bit Parity dataset

3-bit Parity	ABC	ACO	BBO	ES	GA	GWO	PBIL	PSO	TSA
Mean	1.49E-13	2.74E-05	1.44E-13	2.00E-05	6.77E-10	1.39E-09	1.13E-09	3.34E-07	3.87E-09
Best	1.32E-13	5.41E-08	1.44E-13	2.43E-09	1.40E-13	1.34E-13	6.04E-13	7.00E-12	4.01E-13
Worst	2.24E-13	0.00017	1.44E-13	2.99E-05	1.41E-09	2.67E-09	1.73E-08	9.13E-06	2.44E-08
SD	2.54E-14	4.03E-05	2.57E-29	1.26E-05	6.99E-10	1.12E-09	3.51E-09	1.66E-06	5.78E-09
Median	1.37E-13	7.79E-06	1.44E-13	2.99E-05	3.58E-10	1.40E-09	3.42E-12	1.43E-08	2.07E-09
Mean time	1.9296	2.3495	2.0153	2.0680	2.5710	2.2769	1.0279	2.7993	4.0024
Friedman rank	1.6	8.5	2.2	8.4	3.6	4.5	4.2	6.6	5.4
Manuel rank	1	9	4	8	3	2	6	7	5
Wilcoxon	1.73E-06	1.73E-06	1.73E-06	1.73E-06	4.11E-03	6.27E-02	6.64E-04	3.32E-04	0.00E+00
Classification rate (%)	50.00	62.50	50.00	62.50	50.00	50.00	62.50	62.50	50.00

The experimental results for the 3-bit Parity dataset are given in Table 8. ABC is the best in terms of mean training error results. But the classification rate of ABC is 50%. ACO, ES, PBIL, and PSO have the same classification rate (62.5%). The best trained model cannot produce the best classification accuracy.

The experimental results for the 4-bit Encoder Decoder dataset are given in Table 9. All algorithms are trapped in

the same local minima, and all of them produce the same classification rates. Thus, the 4-bit Encoder Decoder is not an identifier problem. This problem has four output values; therefore, our model cannot be appropriate for solving the 4-bit Encoder Decoder problem.

The experimental results for the 3-bits XOR dataset are given in Table 10. GA is the best in terms of mean training

Table 9 Experimental results of the 4-bit Encoder Decoder dataset

4-bit Encoder Decoder	ABC	ACO	BBO	ES	GA	GWO	PBIL	PSO	TSA
Mean	6.23E-02	6.23E-02	6.23E-02	6.25E-02	6.23E-02	6.23E-02	6.23E-02	6.23E-02	6.23E-02
Best	6.23E-02	6.23E-02	6.23E-02	6.23E-02	6.23E-02	6.23E-02	6.23E-02	6.23E-02	6.23E-02
Worst	6.23E-02	6.24E-02	6.23E-02	6.37E-02	6.23E-02	6.23E-02	6.23E-02	6.23E-02	6.23E-02
SD	1.09E-07	2.31E-05	2.82E-17	3.76E-04	1.40E-05	1.60E-07	1.18E-05	3.85E-06	2.32E-06
Median	6.23E-02	6.23E-02	6.23E-02	6.24E-02	6.23E-02	6.23E-02	6.23E-02	6.23E-02	6.23E-02
Mean time	2.2708	2.9314	2.4063	2.4090	2.7637	2.7607	1.1166	3.1452	4.7394
Friedman rank	2.8	8.0	1.5	8.5	5.2	2.3	7.3	5.3	4.1
Manuel rank	1	1	1	1	1	1	1	1	1
Wilcoxon	3.41E-05	1.73E-06	1.73E-06	1.73E-06	2.41E-03	9.71E-05	1.73E-06	1.04E-03	0.00E+00
Classification rate (%)	25	25	25	25	25	25	25	25	25

Table 10 Experimental results of the 3-bits XOR dataset

3-bits XOR	ABC	ACO	BBO	ES	GA	GWO	PBIL	PSO	TSA
Mean	2.97E-17	2.63E-06	3.14E-30	9.28E-10	7.43E-20	7.11E-15	3.84E-12	7.37E-10	1.15E-10
Best	5.93E-20	2.41E-09	3.14E-30	3.40E-15	1.36E-30	3.68E-26	4.83E-19	4.81E-13	1.94E-18
Worst	9.91E-17	2.09E-05	3.14E-30	1.38E-09	2.23E-18	1.35E-13	1.13E-10	1.37E-08	2.75E-09
SD	2.76E-17	4.62E-06	2.14E-45	3.99E-10	4.07E-19	2.78E-14	2.06E-11	2.53E-09	5.06E-10
Median	1.70E-17	5.28E-07	3.14E-30	6.91E-10	1.88E-25	5.64E-22	1.44E-15	4.22E-11	8.68E-14
Mean time	3.1274	4.3180	3.4139	3.4170	3.5980	4.1621	1.3527	3.8864	7.0275
Friedman rank	3.9	9.0	1.1	7.8	2.0	3.4	5.1	7.0	5.7
Manuel rank	4	9	2	7	1	3	5	8	6
Wilcoxon	2.35E-06	1.73E-06	1.73E-06	3.41E-05	1.73E-06	5.31E-05	1.48E-02	1.96E-03	0.00E+00
Classification rate (%)	100	100	100	100	100	100	100	100	100

Table 11 Experimental results of the Sigmoid dataset

Sigmoid	ABC	ACO	BBO	ES	GA	GWO	PBIL	PSO	TSA
Mean	2.46E-01	2.48E-01	2.46E-01	2.49E-01	2.47E-01	2.47E-01	2.47E-01	2.47E-01	2.47E-01
Best	2.46E-01	2.47E-01	2.46E-01	2.47E-01	2.46E-01	2.46E-01	2.47E-01	2.47E-01	2.47E-01
Worst	2.46E-01	2.51E-01	2.46E-01	2.53E-01	2.49E-01	2.47E-01	2.48E-01	2.48E-01	2.48E-01
SD	2.15E-05	1.13E-03	8.47E-17	1.20E-03	5.79E-04	1.67E-04	3.04E-04	3.26E-04	3.47E-04
Median	2.46E-01	2.48E-01	2.46E-01	2.49E-01	2.47E-01	2.46E-01	2.47E-01	2.47E-01	2.47E-01
Mean time	95.9492	34.0330	35.7778	31.2610	31.3496	35.0565	23.0235	47.3332	39.8589
Friedman rank	1.1	7.9	2.6	8.6	4.7	2.8	5.5	5.8	6.2
Manuel rank	1	2	1	2	1	1	2	2	2
Wilcoxon	1.73E-06	2.60E-05	1.73E-06	2.60E-06	5.67E-03	3.18E-06	1.75E-02	2.99E-01	0.00E+00
Classification rate (%)	100.00	94.21	100.00	100.00	100.00	100.00	100.00	100.00	100.00

error results. The classification rate is 100% for all methods. Therefore, 3-bits XOR is not an identifier problem.

The experimental results for the Sigmoid dataset are given in Table 11. ABC, BBO, GA, and GWO share the first position in terms of mean training error results. The classification rate is 100% for all methods except ACO. Therefore, Sigmoid is not an identifier problem.

The experimental results for the Cosine dataset are given in Table 12. GWO and ABC share the best position in terms of mean training error results. PSO is the best in terms of the classification. Cosine is an identifier problem because every algorithm produces different results.

The experimental results for the Sine dataset are given in Table 13. ABC is in the best position in terms of mean training error results. GA is the best in terms of the classification. Sine



Table 12 Experimental results of the Cosine dataset

Cosine	ABC	ACO	BBO	ES	GA	GWO	PBIL	PSO	TSA
Mean	$1.77E-01$	$1.86E-01$	$1.78E-01$	$1.95E-01$	$1.79E-01$	$1.76E-01$	$1.81E-01$	$1.82E-01$	$1.81E-01$
Best	$1.76E-01$	$1.79E-01$	$1.78E-01$	$1.82E-01$	$1.77E-01$	$1.76E-01$	$1.78E-01$	$1.78E-01$	$1.78E-01$
Worst	$1.77E-01$	$1.97E-01$	$1.78E-01$	$2.13E-01$	$1.83E-01$	$1.77E-01$	$1.85E-01$	$1.85E-01$	$1.83E-01$
SD	$3.48E-04$	$4.46E-03$	$5.65E-17$	$8.32E-03$	$1.51E-03$	$4.70E-04$	$1.82E-03$	$1.53E-03$	$1.34E-03$
Median	$1.76E-01$	$1.86E-01$	$1.78E-01$	$1.95E-01$	$1.78E-01$	$1.76E-01$	$1.81E-01$	$1.81E-01$	$1.81E-01$
Mean time	88.9853	29.4610	31.5415	26.7391	26.9503	30.9972	18.8338	43.6489	33.0181
Friedman rank	1.7	7.7	3.6	8.7	3.7	1.3	5.7	6.6	5.9
Manuel rank	1	4	3	5	2	1	3	3	3
Wilcoxon	$1.73E-06$	$1.24E-05$	$2.13E-06$	$1.73E-06$	$1.92E-06$	$1.73E-06$	$9.26E-01$	$1.48E-02$	$0.00E+00$
Classification rate (%)	97.37	78.95	92.11	76.32	94.74	97.37	92.11	100.00	84.21

Table 13 Experimental results of the Sine dataset

Sine	ABC	ACO	BBO	ES	GA	GWO	PBIL	PSO	TSA
Mean	$4.13E-01$	$4.52E-01$	$4.41E-01$	$4.49E-01$	$4.54E-01$	$4.53E-01$	$4.35E-01$	$4.40E-01$	$4.46E-01$
Best	$4.00E-01$	$4.43E-01$	$4.41E-01$	$4.30E-01$	$4.45E-01$	$4.44E-01$	$4.20E-01$	$4.22E-01$	$4.34E-01$
Worst	$4.27E-01$	$4.56E-01$	$4.41E-01$	$4.56E-01$	$4.60E-01$	$4.56E-01$	$4.47E-01$	$4.52E-01$	$4.53E-01$
SD	$6.20E-03$	$4.12E-03$	$1.13E-16$	$7.51E-03$	$3.99E-03$	$1.86E-03$	$6.38E-03$	$7.42E-03$	$5.31E-03$
Median	$4.14E-01$	$4.54E-01$	$4.41E-01$	$4.52E-01$	$4.55E-01$	$4.53E-01$	$4.35E-01$	$4.42E-01$	$4.48E-01$
Mean time	88.4847	36.3617	38.3013	33.9895	34.4315	36.8322	26.4933	48.3780	42.7450
Friedman rank	1.0	7.0	3.6	6.4	8.1	7.2	2.7	3.8	5.1
Manuel rank	1	7	6	4	9	8	2	3	5
Wilcoxon	$1.73E-06$	$2.61E-04$	$1.15E-04$	$7.52E-02$	$2.16E-05$	$2.60E-06$	$1.64E-05$	$3.38E-03$	$0.00E+00$
Classification rate (%)	59.52	57.54	56.75	60.32	67.06	56.75	66.67	60.71	57.94

Table 14 Experimental results of the Balloon dataset

Balloon	ABC	ACO	BBO	ES	GA	GWO	PBIL	PSO	TSA
Mean	$6.22E-17$	$4.13E-07$	$2.22E-31$	$8.22E-10$	$3.15E-23$	$7.44E-19$	$1.65E-13$	$1.33E-10$	$8.46E-12$
Best	$1.39E-20$	$3.99E-10$	$2.22E-31$	$4.68E-14$	$3.40E-40$	$1.10E-31$	$4.10E-21$	$1.43E-15$	$2.08E-17$
Worst	$5.68E-16$	$4.51E-06$	$2.22E-31$	$1.55E-09$	$9.45E-22$	$8.07E-18$	$1.87E-12$	$1.19E-09$	$1.73E-10$
SD	$1.09E-16$	$8.96E-07$	$4.45E-47$	$3.86E-10$	$1.72E-22$	$2.13E-18$	$4.45E-13$	$2.44E-10$	$3.25E-11$
Median	$3.62E-17$	$8.62E-08$	$2.22E-31$	$1.03E-09$	$9.86E-34$	$1.05E-22$	$2.55E-15$	$3.11E-11$	$6.73E-14$
Mean time	10.3630	6.2161	4.8502	4.9285	4.5307	6.0645	1.9197	4.9773	12.5299
Friedman rank	4.3	8.9	1.8	8.0	1.3	2.9	5.1	6.9	5.8
Manuel rank	5	9	3	8	1	2	4	7	6
Wilcoxon	$2.35E-06$	$1.73E-06$	$1.73E-06$	$1.92E-06$	$1.73E-06$	$1.73E-06$	$6.04E-03$	$1.97E-05$	$0.00E+00$
Classification rate (%)	50	0	50	50	50	50	50	50	0

is an identifier problem because every algorithm produces different results.

The experimental results for the Balloon dataset are given in Table 14. GA is in the best position in terms of mean training error results. The best trained model of ACO and TSA cannot classify the test data. ABC, BBO, ES, GWO, PBIL, and PSO have a 50% classification rate. Balloon is an identifier

problem because every algorithm produces different results.

The experimental results for the Iris dataset are given in Table 15. All algorithms except ACO and GA trapped in the same local minima. ABC is the best in terms of the classification. This problem has three output values; therefore, our model cannot appropriate for solving the Iris problem.

Table 15 Experimental results of the Iris dataset

Iris	ABC	ACO	BBO	ES	GA	GWO	PBIL	PSO	TSA
Mean	2.50E-01	2.84E-01	2.50E-01	2.51E-01	3.52E-01	2.50E-01	2.50E-01	2.51E-01	2.52E-01
Best	2.50E-01	2.54E-01	2.50E-01	2.50E-01	2.54E-01	2.50E-01	2.50E-01	2.50E-01	2.50E-01
Worst	2.51E-01	3.32E-01	2.50E-01	2.54E-01	4.91E-01	2.50E-01	2.51E-01	2.55E-01	2.56E-01
SD	1.64E-04	2.02E-02	1.69E-16	8.61E-04	7.18E-02	2.90E-05	1.43E-04	1.03E-03	1.58E-03
Median	2.50E-01	2.84E-01	2.50E-01	2.50E-01	3.19E-01	2.50E-01	2.50E-01	2.51E-01	2.51E-01
Mean time	255.9866	37.5246	42.3152	33.8377	33.6497	41.2012	20.9516	49.5602	56.7811
Friedman rank	3.5	8.2	1.5	5.0	8.8	1.8	3.6	6.2	6.5
Manuel rank	1	2	1	1	2	1	1	1	1
Wilcoxon	5.22E-06	1.92E-06	1.73E-06	4.86E-05	1.73E-06	1.73E-06	2.35E-06	8.97E-02	0.00E+00
Classification rate (%)	20.67	5.33	12.00	8.67	3.33	11.33	10.00	4.67	20.00

Table 16 Experimental results of the Breast Cancer dataset

Breast Cancer	ABC	ACO	BBO	ES	GA	GWO	PBIL	PSO	TSA
Mean	9.36E-03	1.10E-02	2.35E-03	3.76E-02	1.82E-03	1.34E-03	2.55E-02	2.76E-02	2.92E-02
Best	3.95E-03	9.28E-03	2.35E-03	3.49E-02	1.18E-03	1.14E-03	1.24E-02	2.11E-02	1.67E-02
Worst	1.56E-02	1.41E-02	2.35E-03	4.07E-02	8.17E-03	1.55E-03	3.28E-02	3.49E-02	3.39E-02
SD	2.89E-03	1.96E-03	8.82E-19	1.59E-03	1.32E-03	1.26E-04	4.92E-03	3.18E-03	3.45E-03
Median	9.23E-03	9.64E-03	2.35E-03	3.74E-02	1.46E-03	1.35E-03	2.62E-02	2.76E-02	2.96E-02
Mean time	592.9885	178.6869	200.6069	181.4763	170.7403	208.1356	145.6965	189.3432	232.0039
Friedman rank	4.3	4.7	2.9	9.0	1.8	1.4	6.8	6.8	7.4
Manuel rank	4	5	3	9	2	1	6	8	7
Wilcoxon	1.73E-06	1.73E-06	1.73E-06	1.73E-06	1.73E-06	1.73E-06	7.27E-03	2.18E-02	0.00E+00
Classification rate (%)	0	100	100	95	0	0	52	1	77

Table 17 Experimental results of the Heart dataset

Heart	ABC	ACO	BBO	ES	GA	GWO	PBIL	PSO	TSA
Mean	6.49E-25	1.57E-17	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Best	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Worst	1.94E-23	4.70E-16	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
SD	3.55E-24	8.59E-17	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Median	1.12E-33	4.48E-33	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Mean time	562.8594	462.9802	636.0618	466.6571	457.8536	632.8333	225.8350	494.0318	356.5683
Friedman rank	6.7	6.8	4.5	4.5	4.5	4.5	4.5	4.5	4.5
Manuel rank	1	1	1	1	1	1	1	1	1
Wilcoxon	6.10E-05	6.10E-05	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	0.00E+00
Classification rate (%)	100	0	100	100	59.09	100	0	0	100

The experimental results for the Breast Cancer dataset are given in Table 16. GWO is the best in terms of mean training error results, but its trained model cannot classify the test data. ACO and BBO share the best position in terms of the classification. According to these results, it has seen that breast cancer is a challenging problem.

The experimental results for the Heart dataset are given in Table 17. All algorithms achieved the zero error, but ACO, PBIL, and PSO cannot classify the test data. The classification rate of GA is 59.09%. ABC, BBO, ES, GWO, and TSA classify the test data successfully.



Table 18 The manual ranks overview

Manual ranks	ABC	ACO	BBO	ES	GA	GWO	PBIL	PSO	TSA
XOR6	2	3	4	2	3	1	3	1	1
XOR9	1	4	1	3	1	1	1	2	2
XOR13	1	5	1	6	1	1	2	4	3
3-bit Parity	1	9	4	8	3	2	6	7	5
4-bit Enc. Dec	1	1	1	1	1	1	1	1	1
3-bits XOR	4	9	2	7	1	3	5	8	6
Sigmoid	1	2	1	2	1	1	2	2	2
Cosine	1	4	3	5	2	1	3	3	3
Sine	1	7	6	4	9	8	2	3	5
Balloon	5	9	3	8	1	2	4	7	6
Iris	1	2	1	1	1	1	1	1	1
Breast cancer	4	5	3	9	2	1	6	8	7
Heart	1	1	1	1	1	1	1	1	1
Total ranks	24	61	31	57	27	24	37	48	43

Table 19 The Friedman ranks overview

Friedman ranks	ABC	ACO	BBO	ES	GA	GWO	PBIL	PSO	TSA
XOR6	8.4	7.8	6.6	5.8	5.5	1.9	4.7	2.3	1.9
XOR9	3.9	8.4	1.7	8.2	1.7	4.6	4.0	6.8	5.6
XOR13	2.8	8.3	1.4	8.1	3.5	4.6	5.2	5.8	5.3
3-bit Parity	1.6	8.5	2.2	8.4	3.6	4.5	4.2	6.6	5.4
4-bit Enc. Dec	2.8	8.0	1.5	8.5	5.2	2.3	7.3	5.3	4.1
3-bits XOR	3.9	9.0	1.1	7.8	2.0	3.4	5.1	7.0	5.7
Sigmoid	1.1	7.9	2.6	8.6	4.7	2.8	5.5	5.8	6.2
Cosine	1.7	7.7	3.6	8.7	3.7	1.3	5.7	6.6	5.9
Sine	1.0	7.0	3.6	6.4	8.1	7.2	2.7	3.8	5.1
Balloon	4.3	8.9	1.8	8.0	1.3	2.9	5.1	6.9	5.8
Iris	3.3	8.6	1.0	8.1	4.0	2.0	7.1	5.9	5.0
Breast cancer	4.3	4.7	2.9	9.0	1.8	1.4	6.8	6.8	7.4
Heart	6.7	6.8	4.5	4.5	4.5	4.5	4.5	4.5	4.5
Total ranks	45.9	101.6	34.4	100.1	49.6	43.5	68.0	74.0	68.1
General rank	3	9	1	8	4	2	5	7	6

Table 20 The classification rates overview

	ABC	ACO	BBO	ES	GA	GWO	PBIL	PSO	TSA
XOR6	100	100	100	100	100	100	100	100	100
XOR9	100	100	100	100	100	100	100	100	100
XOR13	100	100	100	100	100	100	100	100	100
3-bit Parity	50.00	62.50	50.00	62.50	50.00	50.00	62.50	62.50	50.00
4-bit Enc. Dec	25	25	25	25	25	25	25	25	25
3-bits XOR	100	100	100	100	100	100	100	100	100
Sigmoid	100.00	94.21	100.00	100.00	100.00	100.00	100.00	100.00	100.00
Cosine	97.37	78.95	92.11	76.32	94.74	97.37	92.11	100.00	84.21
Sine	59.52	57.54	56.75	60.32	67.06	56.75	66.67	60.71	57.94
Balloon	50	0	50	50	50	50	50	50	0
Iris	20.67	5.33	12.00	8.67	3.33	11.33	10.00	4.67	20.00
Breast cancer	0	100	100	95	0	0	52	1	77
Heart	100	0	100	100	59.0909	100	0	0	100
Mean CR	69.4	63.3	75.8	75.2	65.3	68.5	66.0	61.8	70.3



Table 21 Experimental results of the Balloon dataset for TSA

Balloon	$N = 10$ ST = 0.1	$N = 10$ ST = 0.5	$N = 10$ ST = 0.9	$N = 50$ ST = 0.1	$N = 50$ ST = 0.5	$N = 50$ ST = 0.9	$N = 100$ ST = 0.1	$N = 100$ ST = 0.5	$N = 100$ ST = 0.9
Mean	4.08E-10	1.26E-10	7.19E-11	5.27E-09	5.17E-09	5.97E-09	2.67E-08	1.71E-08	2.62E-08
Best	9.43E-19	9.81E-21	4.95E-20	1.51E-16	7.38E-14	2.79E-15	3.75E-14	1.01E-14	5.59E-14
Worst	6.51E-09	3.23E-09	1.95E-09	7.56E-08	3.41E-08	1.57E-07	3.29E-07	2.01E-07	1.99E-07
SD	1.47E-09	5.91E-10	3.56E-10	1.44E-08	1.01E-08	2.86E-08	6.56E-08	4.31E-08	5.94E-08
Median	8.42E-14	4.81E-15	8.00E-14	4.06E-10	6.68E-10	5.13E-11	2.41E-09	4.89E-10	9.13E-10
Mean time	8.0720	8.0343	8.7865	5.0810	4.9332	4.7059	4.6307	4.5278	4.2975
Friedman rank	2.8	2.4	2.5	5.9	6.2	5.1	7.3	6.5	6.3
Manuel rank	3	1	2	4	9	5	7	6	8
Wilcoxon	0.00E+00	2.54E-01	4.05E-01	8.19E-05	1.80E-05	6.84E-03	9.32E-06	1.24E-05	1.74E-04
Classification rate %	50.00	50.00	50.00	50.00	50.00	50.00	50.00	0.00	50.00

Table 22 Experimental results of the Iris dataset for TSA

Iris	$N = 10$ ST = 0.1	$N = 10$ ST = 0.5	$N = 10$ ST = 0.9	$N = 50$ ST = 0.1	$N = 50$ ST = 0.5	$N = 50$ ST = 0.9	$N = 100$ ST = 0.1	$N = 100$ ST = 0.5	$N = 100$ ST = 0.9
Mean	2.51E-01	2.51E-01	2.51E-01	2.55E-01	2.54E-01	2.53E-01	2.54E-01	2.53E-01	2.53E-01
Best	2.50E-01	2.50E-01	2.50E-01	2.51E-01	2.50E-01	2.50E-01	2.50E-01	2.50E-01	2.50E-01
Worst	2.54E-01	2.56E-01	2.55E-01	2.69E-01	2.69E-01	2.66E-01	2.76E-01	2.64E-01	2.58E-01
SD	8.92E-04	1.15E-03	1.23E-03	5.24E-03	3.94E-03	3.42E-03	4.71E-03	3.30E-03	2.25E-03
Median	2.50E-01	2.50E-01	2.50E-01	2.52E-01	2.52E-01	2.52E-01	2.53E-01	2.52E-01	2.52E-01
Mean time	122.8270	122.7315	107.2986	42.7832	41.5848	41.1891	31.9500	31.3707	30.3779
Friedman rank	2.6	2.8	3.0	6.5	5.9	6.1	6.3	5.8	5.9
Manuel rank	1	1	1	2	1	1	1	1	1
Wilcoxon	0.00E+00	6.58E-01	7.19E-01	1.97E-05	2.16E-05	3.72E-05	4.29E-06	4.86E-05	1.64E-05
Classification rate %	16.67	27.33	9.33	30.00	20.67	7.33	14.00	30.67	4.00

Table 23 Experimental results of the Cancer dataset for TSA

Cancer	$N = 10$ ST = 0.1	$N = 10$ ST = 0.5	$N = 10$ ST = 0.9	$N = 50$ ST = 0.1	$N = 50$ ST = 0.5	$N = 50$ ST = 0.9	$N = 100$ ST = 0.1	$N = 100$ ST = 0.5	$N = 100$ ST = 0.9
Mean	2.55E-02	2.49E-02	2.54E-02	3.05E-02	2.89E-02	2.93E-02	3.03E-02	3.08E-02	3.00E-02
Best	1.62E-02	1.50E-02	1.25E-02	2.44E-02	1.51E-02	2.43E-02	2.24E-02	2.17E-02	2.19E-02
Worst	3.39E-02	3.44E-02	3.31E-02	3.47E-02	3.41E-02	3.44E-02	3.59E-02	3.62E-02	3.66E-02
SD	4.25E-03	4.32E-03	5.16E-03	2.63E-03	4.34E-03	3.02E-03	3.64E-03	3.61E-03	3.69E-03
Median	2.58E-02	2.54E-02	2.60E-02	3.04E-02	2.98E-02	2.91E-02	3.08E-02	3.21E-02	3.06E-02
Mean time	442.0449	337.4356	362.2962	231.0281	253.7927	311.0903	217.8359	213.5434	227.1923
Friedman rank	3.3	2.9	3.7	6.4	5.5	5.3	5.9	6.1	6.0
Manuel rank	4	2	1	9	3	8	7	5	6
Wilcoxon	0.00E+00	5.58E-01	8.61E-01	3.41E-05	1.71E-03	1.38E-03	1.15E-04	1.15E-04	3.06E-04
Classification rate %	77.00	74.00	0.00	89.00	77.00	81.00	76.00	77.00	87.00



Table 24 Experimental results of the 3-bit parity dataset for TSA

3-bit Parity	$N = 10$ ST = 0.1	$N = 10$ ST = 0.5	$N = 10$ ST = 0.9	$N = 50$ ST = 0.1	$N = 50$ ST = 0.5	$N = 50$ ST = 0.9	$N = 100$ ST = 0.1	$N = 100$ ST = 0.5	$N = 100$ ST = 0.9
Mean	$1.70E-09$	$1.99E-06$	$2.06E-08$	$4.60E-06$	$1.66E-06$	$8.88E-07$	$1.39E-05$	$8.09E-06$	$6.14E-06$
Best	6.41E-13	$8.06E-13$	$8.19E-13$	$1.17E-11$	$1.54E-11$	$7.66E-11$	$5.49E-11$	$2.68E-09$	$4.83E-12$
Worst	$5.34E-09$	$5.95E-05$	$5.28E-07$	$5.17E-05$	$1.44E-05$	$1.89E-05$	$7.07E-05$	$8.19E-05$	$5.04E-05$
SD	$1.85E-09$	$1.09E-05$	$9.60E-08$	$1.15E-05$	$3.29E-06$	$3.54E-06$	$2.18E-05$	$1.85E-05$	$1.29E-05$
Median	$1.26E-09$	$2.52E-09$	$2.56E-09$	$2.28E-08$	$8.53E-08$	$1.66E-08$	$1.76E-06$	$4.26E-07$	$4.15E-07$
Mean time	1.8920	1.8433	1.8087	1.2872	1.2497	1.1923	1.2163	1.2349	1.1735
Friedman rank	2.2	2.8	2.8	5.5	5.5	5.2	7.2	7.0	6.9
Manuel rank	1	2	3	5	6	8	7	9	4
Wilcoxon	$0.00E+00$	$1.59E-01$	$5.71E-02$	$4.29E-06$	$6.34E-06$	$6.98E-06$	$2.13E-06$	$1.73E-06$	$2.13E-06$
Classification rate %	50.00	75.00	50.00	75.00	50.00	50.00	50.00	62.50	75.00

Table 25 Experimental results of the 4-bit Encoder Decoder dataset for TSA

4-bit Encoder Decoder	$N = 10$ ST = 0.1	$N = 10$ ST = 0.5	$N = 10$ ST = 0.9	$N = 50$ ST = 0.1	$N = 50$ ST = 0.5	$N = 50$ ST = 0.9	$N = 100$ ST = 0.1	$N = 100$ ST = 0.5	$N = 100$ ST = 0.9
Mean	$2.06E-02$	$2.05E-02$	$2.06E-02$	$2.05E-02$	$2.06E-02$	$2.05E-02$	$2.07E-02$	$2.06E-02$	$2.06E-02$
Best	2.05E-02	2.05E-02	2.05E-02	2.05E-02	2.05E-02	2.05E-02	2.05E-02	2.05E-02	2.05E-02
Worst	$2.23E-02$	$2.11E-02$	$2.17E-02$	$2.09E-02$	$2.18E-02$	$2.06E-02$	$2.20E-02$	$2.15E-02$	$2.10E-02$
SD	$3.51E-04$	$1.11E-04$	$2.65E-04$	$8.37E-05$	$2.64E-04$	$1.42E-05$	$3.22E-04$	$1.85E-04$	$1.16E-04$
Median	$2.05E-02$	$2.05E-02$	$2.05E-02$	$2.05E-02$	$2.05E-02$	$2.05E-02$	$2.05E-02$	$2.05E-02$	$2.05E-02$
Mean time	2.5172	2.3700	2.3480	1.8992	1.8609	1.7707	1.8014	1.7484	1.7475
Friedman rank	3.6	3.1	3.4	5.4	5.2	5.1	6.9	6.3	6.0
Manuel rank	1	1	1	1	1	1	1	1	1
Wilcoxon	$0.00E+00$	$1.85E-01$	$6.88E-01$	$3.00E-02$	$3.68E-02$	$3.16E-02$	$8.31E-04$	$3.61E-03$	$9.84E-03$
Classification rate %	25.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00

For overall analysis, the manual ranks overview is given in Table 18, the Friedman ranks overview is given in Table 19, and the classification rates overview is given in Table 20.////

ABC and GWO have the same total rank value in terms of mean training error results. BBO is the best in terms of Friedman rank and classification rank values. Every dataset has a different type of search space. An algorithm that solves one problem well cannot solve another problem. This situation is proved by Wolpert and Macready [27], and its name is no free lunch theorems for optimization. The mean classification rate of the TSA is 70.3%. This value is a compatible result. In these experiments, we use fixed stand sizes (10 and 50) and ST (0.1) for TSA. These two peculiar parameters affect the results of the algorithm. In the next experiment, we analyze the different stand sizes and ST values for identifier datasets (Balloon, Iris, Cancer, Parity, EncDec, Cosine, Sine).

tifier datasets (Balloon, Iris, Cancer, Parity, EncDec, Cosine, Sine).

5.1 The Parameter Adjustment for TSA

In this section, we adjust the peculiar parameters of TSA for the 7 identifier datasets (Balloon, Iris, Cancer, Parity, EncDec, Cosine, and Sine). In experiments, we use 10, 50, and 100 as stand sizes and 0.1, 0.5, and 0.9 as ST parameters. The base method for Wilcoxon signed rank test is $N = 10$ and $ST = 0.1$.

The experimental results of the Balloon dataset for the TSA are given in Table 21. $N = 10$ and $ST = 0.5$ variant is the best in terms of mean training error results. All variants except $N = 100$ and $ST = 0.5$ has the same classification rates (50%).

Table 26 Experimental results of the Cosine dataset for TSA

Cosine	$N = 10$ ST = 0.1	$N = 10$ ST = 0.5	$N = 10$ ST = 0.9	$N = 50$ ST = 0.1	$N = 50$ ST = 0.5	$N = 50$ ST = 0.9	$N = 100$ ST = 0.1	$N = 100$ ST = 0.5	$N = 100$ ST = 0.9
Mean	$1.80E-01$	$1.80E-01$	$1.81E-01$	$1.82E-01$	$1.82E-01$	$1.82E-01$	$1.82E-01$	$1.82E-01$	$1.82E-01$
Best	1.77E-01	1.77E-01	$1.78E-01$	$1.78E-01$	$1.78E-01$	$1.78E-01$	$1.78E-01$	1.77E-01	$1.79E-01$
Worst	$1.83E-01$	$1.83E-01$	$1.86E-01$	$1.89E-01$	$1.85E-01$	$1.86E-01$	$1.87E-01$	$1.87E-01$	$1.85E-01$
SD	$1.56E-03$	$1.64E-03$	$1.88E-03$	$2.13E-03$	$1.73E-03$	$1.98E-03$	$2.46E-03$	$2.12E-03$	$1.69E-03$
Median	$1.80E-01$	$1.80E-01$	$1.81E-01$	$1.82E-01$	$1.82E-01$	$1.82E-01$	$1.82E-01$	$1.82E-01$	$1.82E-01$
Mean time	73.7480	66.3652	60.0203	36.9796	55.8863	50.7422	27.1082	27.2170	25.9935
Friedman rank	3.7	4.0	4.2	5.0	5.7	5.6	5.9	5.2	5.5
Manuel rank	1	1	2	2	2	2	2	1	3
Wilcoxon	$0.00E+00$	$7.81E-01$	$3.39E-01$	$1.04E-02$	$1.20E-03$	$2.96E-03$	$3.32E-04$	$2.30E-02$	$5.32E-03$
Classification rate %	36.84	36.84	100.00	36.84	36.84	100.00	44.74	100.00	36.84

Table 27 Experimental results of the Sine dataset for TSA

SineN =	$N = 10$ ST = 0.1	$N = 10$ ST = 0.5	$N = 10$ ST = 0.9	$N = 50$ ST = 0.1	$N = 50$ ST = 0.5	$N = 50$ ST = 0.9	$N = 100$ ST = 0.1	$N = 100$ ST = 0.5	$N = 100$ ST = 0.9
Mean	$4.44E-01$	$4.43E-01$	$4.40E-01$	$4.44E-01$	$4.44E-01$	$4.44E-01$	$4.46E-01$	$4.45E-01$	$4.43E-01$
Best	$4.29E-01$	$4.23E-01$	$4.23E-01$	4.17E-01	$4.27E-01$	$4.29E-01$	$4.33E-01$	$4.29E-01$	$4.24E-01$
Worst	$4.54E-01$	$4.53E-01$	$4.54E-01$	$4.54E-01$	$4.53E-01$	$4.53E-01$	$4.53E-01$	$4.54E-01$	$4.54E-01$
SD	$6.67E-03$	$7.45E-03$	$8.59E-03$	$8.24E-03$	$6.32E-03$	$6.96E-03$	$6.23E-03$	$6.60E-03$	$8.10E-03$
Median	$4.46E-01$	$4.45E-01$	$4.42E-01$	$4.47E-01$	$4.44E-01$	$4.46E-01$	$4.48E-01$	$4.47E-01$	$4.43E-01$
Mean time	56.8857	56.5048	55.8578	37.4354	54.5648	52.6769	52.1033	48.5141	38.4682
Friedman rank	5.3	4.6	3.9	5.4	4.9	5.1	5.5	5.4	4.9
Manuel rank	5	2	2	1	4	5	6	5	3
Wilcoxon	$0.00E+00$	$4.17E-01$	$7.52E-02$	$9.26E-01$	$8.13E-01$	$9.43E-01$	$2.21E-01$	$4.41E-01$	$6.00E-01$
Classification rate %	100.00	52.78	51.98	55.95	100.00	51.98	100.00	89.68	51.98

Table 28 The Friedman ranks overview for the TSA variants

N =	$N = 10$ ST = 0.1	$N = 10$ ST = 0.5	$N = 10$ ST = 0.9	$N = 50$ ST = 0.1	$N = 50$ ST = 0.5	$N = 50$ ST = 0.9	$N = 100$ ST = 0.1	$N = 100$ ST = 0.5	$N = 100$ ST = 0.9
Balloon	2.8	2.4	2.5	5.9	6.2	5.1	7.3	6.5	6.3
Iris	2.6	2.8	3.0	6.5	5.9	6.1	6.3	5.8	5.9
Cancer	3.3	2.9	3.7	6.4	5.5	5.3	5.9	6.1	6.0
Parity	2.2	2.8	2.8	5.5	5.5	5.2	7.2	7.0	6.9
EncDec	3.6	3.1	3.4	5.4	5.2	5.1	6.9	6.3	6.0
Cosine	3.7	4.0	4.2	5.0	5.7	5.6	5.9	5.2	5.5
Sine	5.3	4.6	3.9	5.4	4.9	5.1	5.5	5.4	4.9
Total FR	23.5	22.6	23.5	40.2	38.9	37.5	45.0	42.4	41.4



Table 29 The classification rates overview for the TSA

	$N = 10$ ST = 0.1	$N = 10$ ST = 0.5	$N = 10$ ST = 0.9	$N = 50$ ST = 0.1	$N = 50$ ST = 0.5	$N = 50$ ST = 0.9	$N = 100$ ST = 0.1	$N = 100$ ST = 0.5	$N = 100$ ST = 0.9
Balloon	50.00	50.00	50.00	50.00	50.00	50.00	50.00	0.00	50.00
Iris	16.67	27.33	9.33	30.00	20.67	7.33	14.00	30.67	4.00
Cancer	77.00	74.00	0.00	89.00	77.00	81.00	76.00	77.00	87.00
Parity	50.00	75.00	50.00	75.00	50.00	50.00	50.00	62.50	75.00
EncDec	25.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00
Cosine	36.84	36.84	100.00	36.84	36.84	100.00	44.74	100.00	36.84
Sine	100.00	52.78	51.98	55.95	100.00	51.98	100.00	89.68	51.98
Total FR	50.8	48.7	40.9	51.7	51.4	52.2	51.4	55.0	47.1

Table 30 The classification rates overview

	ABC	ACO	BBO	ES	GA	GWO	PBIL	PSO	TSA
XOR6	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
XOR9	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
XOR13	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
3-bit Parity	50.00	62.50	50.00	62.50	50.00	50.00	62.50	62.50	62.50
4-bit Enc. Dec	25.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00
3-bits XOR	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
Sigmoid	100.00	94.21	100.00	100.00	100.00	100.00	100.00	100.00	100.00
Cosine	97.37	78.95	92.11	76.32	94.74	97.37	92.11	100.00	100.00
Sine	59.52	57.54	56.75	60.32	67.06	56.75	66.67	60.71	89.68
Balloon	50.00	0.00	50.00	50.00	50.00	50.00	50.00	50.00	0.00
Iris	20.67	5.33	12.00	8.67	3.33	11.33	10.00	4.67	30.67
Breast cancer	0.00	100.00	100.00	95.00	0.00	0.00	52.00	1.00	77.00
Heart	100.00	0.00	100.00	100.00	59.09	100.00	0.00	0.00	100.00
Banknote	66.69	62.76	55.54	60.28	55.54	55.54	55.54	61.95	55.54
Diabetic	51.26	58.12	47.87	51.69	51.26	43.96	54.04	54.74	52.13
Twonorm	84.58	73.61	86.16	85.95	95.95	92.95	89.85	83.64	88.59
Ringnorm	63.24	55.04	60.69	64.18	78.19	69.15	67.64	66.59	63.08
Spambase	42.19	43.06	40.62	42.84	42.84	39.40	41.88	42.21	40.69
Mean CR	67.25	62.01	70.93	71.26	65.17	66.19	64.85	61.83	71.38

The experimental results of the Iris dataset for the TSA are given in Table 22. All variants except $N = 50$ and $ST = 0.1$ variant are trapped in the same local minima in terms of mean training error results. $N = 100$ and $ST = 0.5$ variant is the best in terms of classification rates.

The experimental results of the Cancer dataset for the TSA are given in Table 23. $N = 10$ and $ST = 0.9$ variant is the best in terms of mean training error results. $N = 50$ and $ST = 0.1$ variant is the best in terms of classification rates.

The experimental results of the 3-bit Parity dataset for the TSA are given in Table 24. $N = 10$ and $ST = 0.1$ variant is the best in terms of mean training error results. $N = 10$ and $ST = 0.5$, $N = 50$ and $ST = 0.1$ and $N = 100$ and $ST = 0.9$ variants share the best position in terms of classification rates.

The experimental results of the 4-bit Encoder Decoder dataset for the TSA are given in Table 25. All variants are trapped in the same local minima, and all of them produce the same classification rates.

The experimental results of the Cosine dataset for the TSA are given in Table 26. $N = 10$ and $ST = 0.1$, $N = 10$ and $ST = 0.5$ and $N = 100$ and $ST = 0.5$ variants share the best position in terms of mean training error results. $N = 10$ and $ST = 0.9$, $N = 50$ and $ST = 0.9$ and $N = 100$ and $ST = 0.5$ variants share the best position in terms of classification rates.

The experimental results of the Sine dataset for the TSA are given in Table 27. $N = 50$ and $ST = 0.1$ variant is the best in terms of mean training error results. $N = 10$ and $ST = 0.1$, $N = 50$ and $ST = 0.5$ and $N = 100$ and $ST = 0.1$ variants share the best position in terms of classification rates.



Table 31 The classification rates of the Cancer dataset for the TSA variants

Run No	$N = 10$ ST = 0.1	$N = 10$ ST = 0.5	$N = 10$ ST = 0.9	$N = 50$ ST = 0.1	$N = 50$ ST = 0.5	$N = 50$ ST = 0.9	$N = 100$ ST = 0.1	$N = 100$ ST = 0.5	$N = 100$ ST = 0.9
1	79	91	90	87	90	88	81	88	53
2	74	74	91	85	92	82	83	87	57
3	57	80	74	88	61	86	89	83	25
4	84	81	66	33	86	94	87	77	83
5	79	89	83	88	79	90	84	87	82
6	82	65	83	74	84	95	90	84	84
7	88	36	69	77	83	83	79	48	77
8	84	78	83	92	85	83	84	79	82
9	84	84	90	82	84	81	90	90	83
10	84	88	91	79	76	81	79	96	88
11	94	77	85	89	82	85	80	88	89
12	82	79	59	84	83	82	87	87	87
13	95	89	41	84	85	86	88	84	42
14	87	77	86	92	90	81	85	92	83
15	86	76	73	92	86	78	84	88	83
16	86	78	52	83	88	84	86	82	64
17	56	88	47	82	77	86	85	78	81
18	84	78	87	89	83	88	25	75	82
19	78	81	82	91	42	82	89	77	39
20	78	80	34	85	83	84	84	70	90
21	93	85	80	76	87	92	93	81	81
22	87	86	85	45	85	85	56	81	89
23	83	80	37	86	76	82	13	81	82
24	77	80	69	89	90	91	76	87	83
25	78	82	81	83	79	73	88	79	45
26	67	26	78	87	74	91	86	84	84
27	86	81	0	74	90	81	92	53	80
28	79	83	87	83	80	85	89	85	85
29	67	87	83	81	61	70	87	86	85
30	88	61	84	88	84	90	83	63	63
Mean CR	80.87	77.33	71.67	81.60	80.83	84.63	80.07	80.67	74.37
Rank	3	7	9	2	4	1	6	5	8
Max CR	95	91	91	92	92	95	93	96	90
Min CR	56	26	0	33	42	70	13	48	25

For the overall analysis of the TSA variants, the Friedman ranks overview for the TSA variants is given in Table 28 and the classification rates overview for the TSA variants is given in Table 29.

$N = 10$ and $ST = 0.5$ variant is the best in terms of mean Friedman rank results. $N = 100$ and $ST = 0.5$ variant is the best in terms of classification rate values. The last combined values of the classification rate are given in Table 30.

According to the classification rates in Table 30, TSA is the best classifier in these experiment area. TSA is the best solver on 18 different type datasets in terms of mean classification rates. In this work, the second is ES, the third is BBO, the

fourth is ABC, the fifth is GWO, the sixth is GA, the seventh is PBIL, the eighth is ACO, and the last is PSO.

5.2 The Deep Run Analyses for TSA

In the aforementioned experiments, we use the best trained model for classification the test data. When we look at deeply, the best trained model is not the best in the test phase at every time. The classification rates of the Cancer dataset for the TSA variants are given in Table 31.

The maximum classification rate is 96% when $N = 100$ and $ST = 0.5$. According to these results, for better classifi-



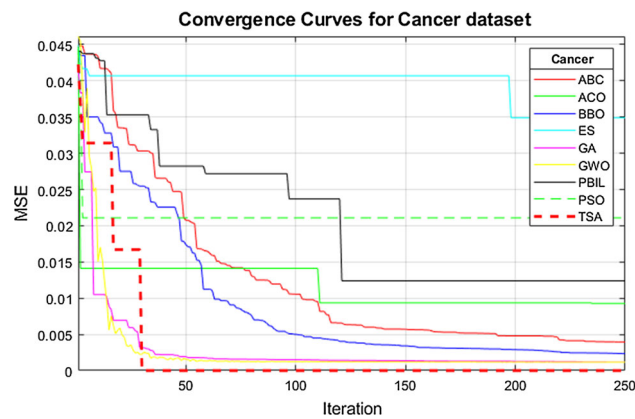


Fig. 4 The convergence graph for the Cancer dataset

cation we should not use only the best trained model, but we must look at the different run model results. The convergence graph for the Cancer dataset is given in Fig. 4.

According to Fig. 4, TSA achieves the best in terms of the mean square error in the training phase for the Cancer dataset. ES, PSO, PBIL, and ACO trapped local minima early but ABC, BBO, GA, GWO, and TSA continue the search process until the termination criterion is met. This graph is also proved the success of the TSA again.

In this type of study, there are two main limitations: general limitations and specific limitations. The general limitations are related to all methods. These are determining the population size; trapping into local optima; detecting the effective exploration and exploitation ratio. The specific limitations of TSA are related to the peculiar parameters of the method. These are determining the search tendency parameter and determining the number of seeds parameter. The search tendency controls the new candidate solution creating scheme, and the number of seeds controls the exploitation in the search space. In this work, we analyzed the search tendency parameter. Experimental results showed that 0.5 is a

good value for the search tendency parameter. This means the new candidate solution creation equations are used as half-and-half.

5.3 The Experiments on Large Datasets

In this section, the experiments on five large datasets are given. The experimental results of Banknote, Diabetic, Twonorm, Ringnorm, and Spambase datasets are located into Tables 32, 33, 34, 35, 36, respectively.

The experimental results for the Banknote dataset are given in Table 32. BBO is in the best position in terms of mean training error results. ABC produced the maximum classification rate.

The experimental results for the Diabetic dataset are given in Table 33. GA is in the best position in terms of mean training error results. ACO produced the maximum classification rate.

The experimental results for the Twonorm dataset are given in Table 34. GA is in the best position in terms of mean training error results. GA produced the maximum classification rate.

The experimental results for the Ringnorm dataset are given in Table 35. GA is in the best position in terms of mean training error results. GA produced the maximum classification rate.

The experimental results for the Spambase dataset are given in Table 36. All algorithms produced the optimum results in terms of mean training error results. ACO produced the maximum classification rate.

6 Conclusion

In this paper, FF MLP ANN is trained by TSA for the first time. TSA is one of the population-based swarm intelligence

Table 32 Experimental results of the Banknote dataset

Banknote	ABC	ACO	BBO	ES	GA	GWO	PBIL	PSO	TSA
Mean	4.39E−20	7.84E−24	1.38E−87	1.02E−53	1.88E−79	1.31E−71	7.72E−69	1.95E−41	6.79E−55
Best	4.81E−34	5.00E−36	1.38E−87	2.37E−70	1.38E−87	1.51E−87	2.95E−87	1.80E−51	1.61E−74
Worst	6.43E−19	1.71E−22	1.38E−87	4.12E−53	4.96E−78	2.04E−70	2.30E−67	5.28E−40	2.04E−53
SD	1.55E−19	3.22E−23	9.08E−103	1.49E−53	9.10E−79	4.79E−71	4.19E−68	9.65E−41	3.72E−54
Median	1.35E−22	2.04E−28	1.38E−87	8.53E−61	1.65E−87	2.43E−79	9.67E−78	1.46E−45	6.19E−66
Mean time	91.9752	5.7195	4.7103	4.7717	4.5926	5.5933	2.8472	4.8154	42.1367
Friedman rank	8.9	8.1	1.0	5.9	2.1	3.1	3.8	7.0	5.1
Manuel rank	8	7	1	5	1	2	3	6	4
Wilcoxon	1.73E−06	1.73E−06	1.73E−06	0.001287	1.73E−06	2.13E−06	1.73E−06	1.73E−06	0
Classification rate (%)	66.69	62.76	55.54	60.28	55.54	55.54	55.54	61.95	55.54

Table 33 Experimental results of the Diabetic dataset

Diabetic	ABC	ACO	BBO	ES	GA	GWO	PBIL	PSO	TSA
Mean	6.81E−02	2.08E−01	2.14E−05	1.52E−01	3.44E−02	7.82E−02	1.39E−01	1.42E−01	1.43E−01
Best	4.07E−03	1.58E−01	2.14E−05	1.21E−01	6.77E−06	3.67E−02	1.06E−01	9.14E−02	6.54E−02
Worst	1.50E−01	2.51E−01	2.14E−05	1.84E−01	1.00E−01	1.31E−01	1.87E−01	1.72E−01	1.76E−01
SD	3.01E−02	2.48E−02	6.89E−21	1.61E−02	2.95E−02	2.20E−02	1.87E−02	2.38E−02	2.29E−02
Median	6.24E−02	2.11E−01	2.14E−05	1.58E−01	3.26E−02	7.77E−02	1.41E−01	1.46E−01	1.44E−01
Mean time	298.6501	57.9424	47.1950	47.5073	37.6225	61.5118	17.7858	38.4946	195.2061
Friedman rank	3.3	8.8	1.1	7.1	2.3	3.5	6.0	6.5	6.4
Manuel rank	3	9	2	8	1	4	7	6	5
Wilcoxon	2.35E−06	1.92E−06	1.73E−06	0.135908	1.73E−06	1.92E−06	0.158855	0.975387	0
Classification rate (%)	51.26	58.12	47.87	51.69	51.26	43.96	54.04	54.74	52.13

Table 34 Experimental results of the Twonorm dataset

Twonorm	ABC	ACO	BBO	ES	GA	GWO	PBIL	PSO	TSA
Mean	1.43E−03	1.27E−01	9.37E−107	3.78E−04	6.10E−59	5.06E−54	3.19E−14	1.50E−06	4.01E−08
Best	1.99E−17	4.18E−02	9.37E−107	1.29E−20	1.10E−139	1.63E−94	4.83E−31	9.20E−18	1.28E−23
Worst	4.30E−02	2.01E−01	9.37E−107	7.26E−03	1.83E−57	1.49E−52	9.34E−13	1.43E−05	1.13E−06
SD	7.85E−03	3.93E−02	2.46E−122	1.48E−03	3.34E−58	2.72E−53	1.70E−13	4.00E−06	2.06E−07
Median	9.37E−17	1.22E−01	9.37E−107	2.87E−08	1.48E−107	4.42E−69	1.18E−21	4.17E−09	3.32E−14
Mean time	2112.1370	97.5102	87.5008	87.7857	77.1834	102.8828	55.6546	77.8253	927.3152
Friedman rank	5.5	9.0	1.5	7.3	1.6	2.9	4.2	7.2	5.7
Manuel rank	8	9	2	6	1	3	4	7	5
Wilcoxon	0.007731	1.73E−06	1.73E−06	0.000529	1.73E−06	1.73E−06	6.32E−05	9.32E−06	0
Classification rate (%)	84.58	73.61	86.16	85.95	95.95	92.95	89.85	83.64	88.59

Table 35 Experimental results of the Ringnorm dataset

Ringnorm	ABC	ACO	BBO	ES	GA	GWO	PBIL	PSO	TSA
Mean	2.13E−02	1.69E−01	6.51E−28	1.03E−01	8.88E−10	5.60E−12	3.18E−02	7.47E−02	5.43E−02
Best	6.21E−15	1.00E−01	6.51E−28	5.52E−02	1.44E−32	6.81E−24	3.46E−07	3.05E−02	2.61E−03
Worst	5.02E−02	2.26E−01	6.51E−28	1.41E−01	2.66E−08	1.48E−10	1.06E−01	1.08E−01	1.12E−01
SD	2.43E−02	3.20E−02	9.12E−44	2.33E−02	4.86E−09	2.72E−11	2.47E−02	2.23E−02	2.72E−02
Median	1.32E−03	1.68E−01	6.51E−28	1.06E−01	3.39E−18	6.39E−17	3.45E−02	7.82E−02	5.05E−02
Mean time	2115.7338	97.4103	87.4961	87.9626	76.6757	102.8440	55.4910	77.9352	928.1096
Friedman rank	4.4	9.0	1.1	7.7	2.3	2.7	5.0	6.9	5.9
Manuel rank	5	3	6	1	7	9	8	4	2
Wilcoxon	4.07E−05	1.73E−06	1.73E−06	3.18E−06	1.73E−06	1.73E−06	0.003854	0.002255	0
Classification rate (%)	63.24	55.04	60.69	64.18	78.19	69.15	67.64	66.59	63.08

algorithms. TSA has two peculiar parameters which are ST and NS. ST controls the exploration and exploitation progress of the algorithm. NS provides better intensification about the current solutions. FF MLP ANN is converted to a vector and TSA optimizes this vector. Eighteen different datasets (XOR6, XOR9, XOR13, 3-bit Parity, 4-bit Encoder Decoder, 3-bits XOR, Sigmoid, Cosine, Sine, Balloon, Iris, Breast Cancer, Heart, Banknote, Diabetic, Twonorm, Ringnorm,

and Spambase) are used in experiments. TSA is compared with PSO, GWO, GA, ACO, ES, PBIL, ABC, and BBO. The experimental results show that TSA is the best in terms of mean classification rates and outperformed the opponents on 18 problems. The obtained results are proven by two different statistical (Wilcoxon signed rank test and Friedman's test) tests. Generally speaking, the swarm-based methods suffer from low exploration, but TSA has an efficient exploration



Table 36 Experimental results of the Spambase dataset

Spambase	ABC	ACO	BBO	ES	GA	GWO	PBIL	PSO	TSA
Mean	0.00E + 00	0.00E + 00	0.00E + 00	0.00E + 00	0.00E + 00	0.00E + 00	0.00E + 00	0.00E + 00	0.00E + 00
Best	0.00E + 00	0.00E + 00	0.00E + 00	0.00E + 00	0.00E + 00	0.00E + 00	0.00E + 00	0.00E + 00	0.00E + 00
Worst	0.00E + 00	0.00E + 00	0.00E + 00	0.00E + 00	0.00E + 00	0.00E + 00	0.00E + 00	0.00E + 00	0.00E + 00
SD	0.00E + 00	0.00E + 00	0.00E + 00	0.00E + 00	0.00E + 00	0.00E + 00	0.00E + 00	0.00E + 00	0.00E + 00
Median	0.00E + 00	0.00E + 00	0.00E + 00	0.00E + 00	0.00E + 00	0.00E + 00	0.00E + 00	0.00E + 00	0.00E + 00
Mean time	2751.0268	468.9487	418.0443	428.8673	344.7158	538.8551	154.8249	341.6761	1754.3480
Friedman rank	1	1	1	1	1	1	1	1	1
Manuel rank	1	1	1	1	1	1	1	1	1
Wilcoxon	1	1	1	1	1	1	1	1	0
Classification rate (%)	42.19	43.06	40.62	42.84	42.84	39.40	41.88	42.21	40.69

mechanism. In future studies, the improved versions of TSA would be used for training FF MLP ANN.

Funding The authors wish to thank Scientific Research Projects Coordinatorship at Selcuk University and The Scientific and Technological Research Council of Turkey for their institutional supports.

Compliance with ethical standards

Conflicts of interest The authors declare that they have no conflict of interest.

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